Emily L. Martin

Irvine, California • elmarti2@uci.edu

PROFESSIONAL INTEREST

PhD Candidate in Ecology and Evolutionary Biology pursuing an academic career as both a researcher and educator, with a focus on freshwater conservation and applied science

EDUCATION

- **PhD in Ecology and Evolutionary Biology**, University of California - Irvine, Irvine, CA
  Expected May 2027
- **MSc in Biological Sciences**, University of California - Irvine, Irvine, CA
  March 2025
- **BS in Biology**, Purdue University, West Lafayette, Indiana
  May 2022

RESEARCH EXPERIENCE

Graduate Student Researcher, Symons Lab, University of California- Irvine, 2022- Present
Science Advisor, Board of Advisors, Crowley Fish Foundation, 2025- Present
Visiting Independent Researcher, Derry Lab, *Université du Québec à Montréal,* 2023
Undergraduate Researcher, Searle & Christie Labs, Purdue University, 2019-2022

PUBLICATIONS

1. **Martin, E. L.,** Adarve-Rengifo, I., Siddons, S., Blackwood, P. E., Hua, J., & Searle, C. L. (2026). Leaf litter leachate limits fungal pathogen growth but not amphibian infection. *Ecology and Evolution*, 16(5). https://doi.org/10.1002/ece3.73668

2. Blackwood, P. E., **Martin, E. L.**, & Searle, C. L. (2025). Temperature variability and salt pollution interact to alter subsequent multi-parasite susceptibility in larval amphibians. *Freshwater Biology*, 70(9), e70080. https://doi.org/10.1111/fwb.70080

3. Nalesnik, A. M., **Martin, E. L**., Kovacs, I. S., Johnson, C. S., Carroll, E. I., Jubar, A., Hemstrom, W., Wilkie, M. P., Dunlop, E. S., Sepulveda, M. S., Johnson, N. S., & Christie, M. R. (2025). Larger larval sea lamprey (Petromyzon marinus) have longer survival times when exposed to the lampricide 3-trifluoromethyl-4-nitrophenol. *Journal of Great Lakes Research*, 51(5), 102620. https://doi.org/10.1016/j.jglr.2025.102620

4. Sorte, C. J. B., Beshai, R., Dean, L., Fales, R., **Martin, E. L.**, & Sherza, S. (2024). Evolution and invasive species. In R. M. Kliman (Ed.), *Encyclopedia of Evolutionary Biology* (pp. 282–287). Academic Press. https://doi.org/10.1016/B978-0-12-800049-6.00303-6

IN REVISION OR REVIEW

1. **Martin, E. L**., Fiorenza, E., & Symons, C. C. (in review). Prey functional change exceeds taxonomic change with introduced predators. *Oikos.*

2. Backenstose, N. J. C., Nalesnik, A. M., Tsai, D.-Y., **Martin, E. L.**, Jubar, A. K., Kovacs, I. S., Bruning, T. M., Dunlop, E. S., Sepulveda, M. S., Johnson, N. S., & Christie, M. R. (in review). Pesticide resistance is evolving in invasive sea lamprey. *Nature Communications.*

3. Mohan, J., **Martin, E. L**., & Symons, C. C. (in revision). A persistent aquatic invasive species problem highlights the need for a paleontological perspective: Didymosphenia geminata is not an invasive species in the Northern Hemisphere. *Limnology and Oceanography Letters.*

4. Bates, A., Suding, K., Haddad, N., Mastretta-Yanes, A., McGill, B., Prugh, L., Rangel, R., Sorte, C. J. B., Symons, C. C., **Martin, E. L.,** & 44 contributing authors (in press). Chapter 3: Direct effects of climate change on biodiversity. *USGS Report on Biodiversity and Climate Change.*

5. Bates, A., Haddad, N.,...Martin, E.,...Suding, K., & 48 other contributing authors (submitted). Climate Change Is Restructuring Biodiversity in North America Faster Than Adaptive Responses Can Keep Pace*. Science.*

IN PREPARATION

1. **Martin, E. L.**, Jeong, Y., Ramsey, S., Soriano, M., Mohan, J., & Symons, C. C. (submission 11/6/26; draft available upon request). Hydra establishment alters zooplankton composition and trophic structure in mesocosms. *Journal of Plankton Research*.

2. Mohan, J., Wall, C., Shurin, J., Bonadonna, C., **Martin, E. L.**, ... Symons, C. C. (in preparation; draft available upon request). Insights from the Sierra Nevada Range, California, USA: Drivers of zooplankton community structure in a highly variable environment. *Global Change Biology.*

3. Soriano, M. U., Baker, H. K., Wall, C. B., Dominguez, J. H., **Martin, E. L.**, ... & Symons, C. C. (in preparation; draft available upon request). Climate warming drives thermophilization of zooplankton communities in mountain lakes. G*lobal Change Biology.*

4. Hellwitz, S., Schlauch, J., **Martin, E.L**., …, Symons, C. C. (in preparation; draft available upon request). Reimagining the Undergraduate Ecology and Evolutionary Biology Curriculum. *Bioscience.*

5. Louie, J., Mohan, J., **Martin, E.**, Martinez, C., Symons, C. (in preparation; draft available upon request). Lake-Specific Variation in Brook Trout Morphology Across Sierra Nevada Lakes. *Freshwater Biology.*

PRESENTATIONS

1. Martin, E., Symons, C. (2026, May). *Fish Reduce Secondary Productivity of Zooplankton in Mountain Lakes* (Talk). Association for the Sciences of Limnology and Oceanography (ASLO), Montreal, Quebec, Canada.

2. Martin, E. (2026, April). *Life Beneath the Surface: Zooplankton and Mountain Lake Ecosystems* (Invited Talk). The Ocean Institute Distinguished Speaker Series, Dana Point, CA.

3. Martin, E., Symons, C. (2026, March). *Fish Reduce Secondary Productivity of Zooplankton in Mountain Lakes* (Talk). Winter Ecology & Evolutionary Biology Graduate Symposium, Irvine, CA.

4. Martin, E., Fiorenza, E., Symons, C. C. (2026, February). *Introduced Fish Constrain Zooplankton Functional Diversity in Mountain Lakes* (Talk). Predator–Prey Interactions Gordon Research Conference, Lucca, Italy.

5. Martin, E. (2025, August). *Fish Impacts and Citizen Science on Mountain Lakes* (Talk). Inyo Fish & Game Commission Board Meeting, Bishop, CA.

6. Martin, E., Symons, C. (2025, March). *Introduced Fishes Constrain Zooplankton Functional Diversity in Mountain Lakes* (Talk). Mildred E. Mathias Conference, Bodega Bay, CA.

7. Martin, E., Symons, C. (2024, April). *Introduced Fishes Constrain Functional Diversity in Mountain Lakes* (Poster). Sierra Nevada Symposium, Merced, CA.

8. Martin, E., Symons, C. (2024, March). *Introduced Fishes Constrain Functional Diversity in Mountain Lakes* (Talk). Winter Ecology & Evolutionary Biology Graduate Symposium, Irvine, CA.

9. Martin, E. (2023, June). *A LOREX Perspective: Navigating and Embracing "Newness"* (Talk). Aquatic Sciences Meeting 2023, Mallorca, Spain.

10. Martin, E. (2023, May). *Research Overview and ASLO LOREX Experience* (Talk). Université du Québec à Montréal (UQAM), Montreal, Canada.

11. Martin, E., Symons, C. (2023, March). *Introduced Fishes Constrain Functional Diversity in Alpine Lakes* (Poster). Winter Ecology & Evolutionary Biology Graduate Symposium, Irvine, CA.

12. Martin, E., Blackwood, P., Siddons, S., Searle, C. (2022, November). *Plant Leachates Impact Amphibian Infection with a Fungal Pathogen* (Talk). ASU IRCEB Amphibian Disease Meeting, Virtual.

13. Martin, E. (2022, April). *Plant Leachates Impact Amphibian Infection with a Fungal Pathogen* (Poster). Purdue Spring Undergraduate Research Conference, Virtual.

14. Martin, E. (2021, July). *Plant Leachates Impact Amphibian Infection with a Fungal Pathogen* (Poster). Joint Meeting of Ichthyologists and Herpetologists (JMIH), Virtual.

15. Martin, E. (2021, April). *Plant Leachates Impact Infection Dynamics in an Amphibian Fungal Pathogen* (Poster). Purdue Spring Undergraduate Research Conference, Virtual.

HONORS, GRANTS & AWARDS

- 2026 GAANN Fellowship Recipient ($28,428)
- 2026 ASLO EcoDAS Participant ($3000)
- 2026 GAANN Travel Fund Grant ($2000)
- 2026 Valentine Eastern Sierra Reserve Graduate Award ($1500)
- 2026 GAANN Fellowship Recipient ($17,136)
- 2025 GAANN Travel Fund Grant ($2064)
- 2025 OSyM Quantitative Modeling in Biology Participant ($2000)
- 2025 Valentine Eastern Sierra Reserve Graduate Award ($1500)
- 2025 GAANN Fellowship Recipient ($38,000)
- 2023 Mathias Graduate Student Research Grant UCNRS ($3000)
- 2023 Undergraduate Research Opportunity Program (UROP/Calit2) ($3000)
- 2023 ASLO LOREX Student Fellowship ($6000)
- 2022 UCI Ecology and Evolutionary Biology Excellence Fellowship ($7000)
- 2022 NSF GRFP Honorable Mention
- 2021 Dr. Eric Dwayne Miller Scholarship ($2,000)
- 2021 Purdue University OUR Travel Grant ($125)
- 2021 Dr. William H. Phillips Summer Research Fellowship ($4,000)
- 2021 McAtee Award ($2,500) – declined
- 2021 M and J Singh Scholarship ($1,100)
- 2020 Cable-Silkman Fellowship ($4,000)
- 2020 Richard Bare Biological Sciences Scholarship ($1,000)
- 2019 Summer Stay Scholar – Purdue University ($2,500)
- 2019 Purdue University Big Move Short-Term Scholarship ($500)
- 2018–2022 Purdue University, Department of Biological Sciences Dean's List
- 2018–2021 Purdue University Presidential Scholarship ($16,000)
- 2018–2021 Purdue University Marquis Scholarship ($6,860)

LEADERSHIP & SCIENCE OUTREACH EXPERIENCE

- **Co-Leader,** California Wildfire-Watershed Working Group, Product of EcoDAS collaboration, Honolulu, Hawaii, 2026
- **Fishery Information Booth Leader,** Owens River Cleanup, Bishop, CA, Fall 2025 and 2026
- **Invited Speaker**, Science on Tap for NoHotCreekMine, Mammoth Lakes, CA, June 2026
- **Organizer and Instructor**, Aquatic Ecology Day, Sierra Nevada Aquatic Research Laboratory (SNARL), Mammoth Lakes, CA, September 2025
- **Invited Speaker**, Inyo County Fish and Game Commission Board Meeting, Bishop, CA, August 2025

- **Podcast Guest**, “Tuesday Talks with Fred,” Bishop, CA, August 2025
- **Radio Guest**, Sierra Wave Radio, Bishop, CA, August 2025
- **Science Advisor** for Crowley Fish Foundation, 2025–Present
- **Science Outreach Volunteer**, Native Pollinators Event, Orange County Market Science Event, June 2025
- **Radio Guest**, “Office Hours” with Newton Hood (KUCI), Topic: Mountain Lakes and Freshwater Research, 2025
- **EEB Graduate Student Representative**, Department of Ecology and Evolutionary Biology, University of California, Irvine, 2024–Present
- **Graduate Leadership Team Member** — Department of Ecology and Evolutionary Biology, University of California, Irvine, 2024–2026
- **Leader** for development and implementation of the Annual Graduate Student Survey
- **Panelist**, Sierra Nevada Aquatic Research Laboratory (SNARL), Mammoth Lakes, CA, Summer 2024
- **Winter Symposium and Rooftop Social Coordinator**, Department of Ecology and Evolutionary Biology, University of California, Irvine, Winter 2024
- **Graduate Recruitment Coordinator**, Department of Ecology and Evolutionary Biology, University of California, Irvine, Winter 2024
- **Volunteer**, Newport Beach Cleanup, Newport Beach, CA, Winter 2024
- **Panelist**, Sierra Nevada Aquatic Research Laboratory (SNARL), Mammoth Lakes, CA, Summers 2023, 2024, 2026

TEACHING

- **BIO 90L, Introduction to Biological Research**. University of California, Irvine (Fall 2025 and 2026). Graduate Teaching Assistant, Department of Ecology and Evolutionary Biology. Dr. Ana Garcia-Vedrenne.
- **BIO 93L, Introduction to Biological Research.** University of California, Irvine (September–December 2024). Graduate Teaching Assistant, Department of Ecology and Evolutionary Biology. Dr. Ana Garcia-Vedrenne.
- **BIO 94L, Introduction to Biological Research.** University of California, Irvine (January–March 2024). Graduate Teaching Assistant, Department of Ecology and Evolutionary Biology. Dr. Ana Garcia-Vedrenne.
- **EcoEvo 207, Quantitative Methods.** University of California, Irvine (September–December 2023). Graduate Teaching Assistant, Department of Ecology and Evolutionary Biology. Dr. Amy Henry.
- **BIO 94, Organisms to Ecosystems.** University of California, Irvine (January–March 2023)**.** Graduate Teaching Assistant, Department of Ecology and Evolutionary Biology. Dr. Tobin Hammer and Dr. Chris Martinez.
- **BIO 93, Organisms to Ecosystems.**. University of California, Irvine (September–November 2022). Graduate Teaching Assistant, Department of Ecology and Evolutionary Biology. Dr. Kim Greene, Dr. Marcelo Wood, and Dr. Jorge Busciglio.
- **BIOL 58000, Evolution.** Purdue University (February–May 2021). Teaching Intern, Department of Biological Sciences. Dr. Mark Christie.

MENTORING

1. **Yunsuh (Felix) Jeong (2 years + 1 summer fieldwork)**
   Program: Excellence in Research, Bio199
   Mentor Role: taught skills in macroinvertebrate and zooplankton identification, introduced student to R and statistics, 2024 UROP proposal, 2025 SURP proposal, 2026 thesis, weekly meetings, mentor during field season
   Student Accomplishments: completed an Excellence in Research honors thesis, awarded SURP funding, awarded UROP funding, completed a UROP independent project, learned technical skills for future career
2. **Caitlyn Cargnoni (1 year)**
   Program: Bio199 and Calit2 Program

Mentor Role: taught microscopy skills for learning and taking photos of phytoplankton, fostered brainstorming for the Calit2 project, weekly meetings, aided in statistical analysis for project, provided feedback on multiple drafts, participated and organized YouTube video for Calit2 project
Student Accomplishments: development of microscopy skills and phytoplankton identification, completed and attained a Calit2 grant ($3000), experienced a research project from design to completion, presented a poster at UROP

3. **Rizelle Mazon (1 year)**
   Program: Bio199 and Calit2 Program
   Mentor Role: instructed zooplankton ephippia identification, taught microscopy skills for learning and taking photos of phytoplankton, fostered brainstorming for the Calit2 project, weekly meetings, aided in statistical analysis for project, provided feedback on multiple drafts, participated and organized YouTube video for Calit2 project
   Student Accomplishments: development of microscopy skills and phytoplankton identification, completed and attained a Calit2 grant ($3000), experienced a research project from design to completion, helped create a poster at UROP
4. **Soffia Ramsey (1 year + 1 summer of fieldwork)**
   Program: Bio199
   Mentor Role: helped assist with experimental design and provide feedback for individual research project, taught new field skills over summer 2023
   Student Accomplishments: development of field skills, presented poster at both Sierra Nevada Research Symposium and UROP, manuscript in preparation
5. **Meili Soriano (2 years + 2 summers of fieldwork)**
   Program: Bio199
   Mentor Role: helped assist with experimental design and provide feedback for individual research project, taught new field skills over summer 2023, provided written feedback for various opportunities (UROP, NSF GRFP)
   Student Accomplishments: development of field skills, presented poster at both Sierra Nevada Research Symposium and UROP, manuscript in preparation
6. **Jenny Louie (2 quarters)**
   Program: Bio199
   Mentor Role: assist fish morphometrics through feedback on paper draft, aid on analysis, and show illustration methods; introduce statistics for fishery citizen science data
   Student Accomplishments: presented poster at UROP, drafting paper for publication
7. **Nathan Bogin (1 year+ 1 summer of fieldwork)**
   Program: Bio199
   Mentor Role: taught skills in literature review, Excel, and study design using Sierra fish stocking data organization, taught skills for his first field season
   Student Accomplishments: curated a wide collection of historical fish stocking data
8. **Brandy Van Nguyen (1 quarter + 1 summer of fieldwork)**
   Program: SURP
   Mentor Role: provide feedback for individual research project, taught new field skills over summer 2024
   Student Accomplishments: field sampling techniques, collaboration skills
9. **Ruth Zhou (2 quarters)**
   Program: Bio199
   Mentor Role: taught skills in literature review using fish stocking data organization, introduced data entry strategies for Excel/Google Sheets
   Student Accomplishments: literature review skills and exposure to research
10. **Yuemeng Li (1 quarter)**
    Program: Lab Volunteering
    Mentor Role: showed how to use a microscope and how to count/identify zooplankton resting eggs
    Student Accomplishments: microscopy skills and exposure to research

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PROFESSIONAL AFFILIATIONS

- Ecological Society of America, 2026.
- Society for Conservation Biology, 2026
- Association for the Sciences of Limnology and Oceanography, 2023–Present.
- American Society of Ichthyologists and Herpetologists, 2021.
- Herpetologists' League, 2021.
- Society for the Study of Amphibians and Reptiles, 2021.